Grace Jin

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EDUCATION

Cornell University, College of Engineering — Expected May 2027

B.S. Computer Science, Minor in Artificial Intelligence

- **Relevant Coursework:** Systems Programming, Compilers in C++, Physically Based Rendering, Computer Graphics, Machine Learning, Data Structures and Algorithms, Digital Logic and Computer Organization

EXPERIENCE

Software Engineering Intern | LinkedIn, Mountain View, CA — May 2026 – Present

- Core Systems Performance team, building an LLM-driven linter and autofixer that refactors blocking code to be async/concurrent, reducing production server occupancy.

Software Engineering Intern | Cepton Technologies, San Jose, CA — May 2025 – Aug. 2025

- Built a real-time WGPU-accelerated 3D visualization pipeline on Linux systems to simulate LiDAR data rendered over synthetic autonomous vehicle driving geometries at 60+ FPS
- Implemented WGSL compute shaders to perform parallel raycasting for ML model training data collection
- Built a 3D environment reconstruction system in Rust to generate simulator depth maps integrated within GPU pipeline
- Replaced legacy Rust transformations with library-backed matrix kernels (SVD, affine), reducing compute time by 40%
- Authored engineering blog post documenting this project: developer.cepton.com/blog/AR_simulator

Lead Software Developer | Cornell Center for Teaching Innovation, Ithaca, NY — Oct. 2024 – Present

- Develop 3D Unity visualizations for a 500+ student EM course (grant-funded, with 4000+ playthroughs), and support 30+ students prototyping AR/VR apps on Meta Quest and Snap OS

Software Developer | Cornell People and Robots Teaching and Learning (PoRTaL), Ithaca, NY — Aug. 2024 – May 2025

- Implemented 20+ training tasks with PDDL, improving LLM reasoning with complex and asynchronous tasks
- Engineered a PyGame interface to generate thousand-line JSON specs for an LLM planning benchmark

PROJECTS

Physically-Based Ray Tracer [GitHub] | C++, DirectX, HLSL, DXR — Jan. 2025 – Present

- Built a real-time DXR path tracer with 16-bounce GI, Chiang fiber BCSDF, Disney BSDF, null-collision volumetrics, HDR environment lighting, and balance-heuristic MIS
- Designed a DXR AnyHit-only shadow path with payload Fresnel for transparent and dielectric occluders, skipping closest-hit to cut shader cost
- Accelerated volume rendering with majorant-mip empty-space skipping and analytic fast-path on constant extinction
- Profiled bottlenecks via Nsight Graphics, reducing ray payload and shader register pressure to improve warp occupancy

CUDA Graphics Rasterizer Engine [GitHub] | C++, CUDA, SDL2 — Jul. 2025 – Present

- Built a software rasterizer in C++ implementing the full graphics pipeline with no external graphics libraries
- Ported the rasterizer to CUDA for ∼100x throughput over the CPU baseline by batching per-frame triangle rasterization into single-dispatch kernels, eliminating per-triangle launch overhead
- Profiled via Nsight Compute, coalescing framebuffer writes through tile binning to improve global store throughput

Eta Language Compiler in C++ [GitHub] | C++, Bison, x86, Docker — Jan. 2025 – Present

- Build an end-to-end compiler for Eta, a C-like statically-typed language, spanning lexical analysis, LALR(1) parsing, type checking, IR lowering, and x86-64 code generation with a team of 4

NeuroScent - MIT Reality Hack Hardware Track Winner [DevPost] | C#, Unity, OpenBCI, Arduino — Jan. 2025

- Led team of 5 to develop an immersive VR olfactory biofeedback system and Galea EEG data processor for mental well-being enhancement, won out of 400+ competitors
- Integrated Unity to render calming scenes and trigger Arduino-controlled diffusers upon detecting abnormal biofeedback

Computer Science Content Creator [Instagram] — Aug. 2019 – Present

- Built an audience of 18K+ followers and 3M+ video views by posting computing topics, personal projects and digital art

SKILLS

Programming Languages: C++, Python, Rust, Java, C#, Typescript, JavaScript, SQL

GPU Computing & Graphics: DirectX, HLSL, WGPU/WebGPU, CUDA, WGSL, Vulkan, OpenGL/WebGL

Development Tools: Linux, Git, Nsight Graphics, Nsight Compute, Nsight Systems, Kubernetes, Docker, GCC, GDB, RTOS, Blender, Figma